

Karnatak University Mathematics Teacher's Association (KUMTA)

B. Sc IV Semester (RVSD NEP) Question Bank

Subject: MATHEMATICS Discipline Specific Core Course (DSC) Course

Title: Differential Equations and Integral transforms

Course Code: C4 MAT 1 T1 (Theory)

Type of Course	Theory /Practical	Credits	Instruction hours per week	Total No. of Lecture	Duration of Sem End Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
DSC – 7	Theory	04	04	60 Hrs	03 Hrs	20	80	100

Syllabus

Unit – I Ordinary Differential Equations 15 Hrs.

Bernoulli's differential equations. Exact differential equations: Necessary and sufficient condition for the equations to be exact, Reducible to the exact differential equations. Differential equations of the first order and higher degree: Equations solvable for p, x, y. Clairaut's equation, Linear differential equations of the nth order with constant coefficients. Particular Integrals when the RHS is of the form e^{ax} , $\sin(ax+b)$, $\cos(ax+b)$, x^n , $e^{ax}.V$ and $x.V$ (with proofs), where V is a function of x.

Unit – II Partial Differential Equations 15 Hrs.

Formation of partial differential equations by elimination of arbitrary constants and functions, **Condition of integrability of $Pdx+Qdy+Rdz=0$** , Solution of partial differential equations of first order, Lagrange's linear equations of the form $Pp + Qq = R$, Standard types of first order non-linear partial differential equations. The integrals of the non-linear equation by Charpit's method. Homogeneous linear partial differential equations with constant coefficients.

Unit – III Laplace Transforms 15 Hrs.

Definition, Basic Properties, Laplace transforms of some standard functions, Laplace transform of Periodic functions, Laplace transform of derivative and integral of a function, Heaviside function, Dirac-delta function, Convolution theorem, Inverse Laplace transforms and its properties, Solution of differential equations by using Laplace transforms.

Unit – IV Fourier Series and Transforms 15 Hrs.

Periodic functions, Fourier coefficients, Fourier series of functions with period 2π and period $2L$. Fourier series of even and odd functions, Half range cosine and sine series, Fourier Transforms: Finite Fourier cosine and sine transforms, Transforms of derivatives.

Unit – I Ordinary Differential Equations

I. Questions carrying Two Marks each

1. Define Bernoulli's differential equation .

2. Define Exact differential equation .

3. Solve $\frac{dy}{dx} + \frac{y}{x} = y^2x$

4. Verify whether the following differential equation is exact

(i) $x dy + (x + y)dx = 0$

(ii) $(e^y + 1) \cos x dx + e^x \sin x dy = 0$

(iii) $(x^2 - ay) dx + (y^2 - ax) dy = 0$

(iv) $(a^2 - 2xy - y^2) dx - (x + y)^2 dy = 0$

(v) $(\cos y + y \cos x) dx + (\sin x - x \sin y) dy = 0$

5. Solve (i) $p^2 - 5p + 6 = 0$

(ii) $p^2 - 7p + 12 = 0$

(iii) $p^2 - ax^3 = 0$

(iv) $p^2 - 5p - 6 = 0$

6. Solve $\sin px \cos y - \cos px \sin y = p$

7. Solve (i) $y = px + p^2$ (ii) $y = px + p - p^2$

8. Solve (i) $4 \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + y = 0$ (ii) $2 \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 4y = 0$

(iii) $\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = 0$ (iv) $\frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 13y = 0$

9. Solve (i) $(D^2 + D + 1)y = 0$ (ii) $(D^3 + 1)y = 0$

(iii) $(D^3 + 3D^2 + 3D + 1)y = 0$ (iv) $(9D^2 + 12D + 4)y = 0$

10. Define Clairaut's equation

11. Find the general solution of

(i) $y = px + \frac{a}{p}$

(ii) $y = px + ap^2$

(iii) $y = px - cp$

(iv) $y = px + \sin^{-1}p$

II. Questions carrying Five Marks each

1. Solve

$$(i) (x^3y^2 + xy)dx = dy$$

$$(ii) x \frac{dy}{dx} + y = x^3y^6$$

$$(iii) x \frac{dy}{dx} + y = y^2 \log x$$

$$(iv) x \frac{dy}{dx} + y = y^2x^3 \cos x$$

$$(v) x^3 \frac{dy}{dx} - x^2y + y^4 \cos x = 0$$

2. Prove that the necessary and sufficient conditions for the differential

$$\text{equation } M dx + N dy = 0 \text{ to be exact is } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

3. Solve

$$(i) (x^2y - 2xy^2) dx - (x^2 - 3x^2y) dy = 0$$

$$(ii) x^2y dx - (x^3 + y^2) dy = 0$$

$$(iii) y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0$$

$$(iv) (xy \sin xy + \cos xy) y dx + (xy \sin xy - \cos xy) dy = 0$$

$$(v) (x^2y^2 + xy + 1) y dx + (x^2y^2 - xy + 1) x dy = 0$$

4. Solve

$$(i) \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + 3y = e^{3x}$$

$$(ii) \frac{d^2y}{dx^2} - \frac{dy}{dx} + y = e^{2x} + e^x$$

$$(iii) 6 \frac{d^2y}{dx^2} + 17 \frac{dy}{dx} + 12y = e^{-x}$$

$$(iv) \frac{d^2y}{dx^2} + 6 \frac{dy}{dx} + 9y = 2e^{-3x} + e^x$$

$$(v) \frac{d^2y}{dx^2} + 4y = x \sin x$$

$$(vi) \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + y = x \cos x$$

5. Solve

$$(i) x^2p^2 dx + xyp - 6y^2 = 0$$

$$(ii) y = 2px + p^4x^2$$

$$(iii) \quad y = 2px - p^2$$

$$(iv) \quad y^2 + xyp - x^2p^2 = 0$$

$$(v) \quad xyp^2 - (x^2 + y^2)p + xy = 0$$

$$(vi) \quad xp^2 - 2yp + x = 0$$

$$(vi) \quad yp^2 - 2xp + y = 0$$

$$(vii) \quad p^3 - 2xyp + 4y^2 = 0$$

6. With usual notation prove that $\frac{1}{f(D)} e^{mx} = \frac{1}{f(m)} e^{mx}$, where $f(m) \neq 0$

7. With usual notation prove that $\frac{1}{f(D^2)} \sin ax = \frac{1}{f(-a^2)} \sin ax$,
where $f(-a^2) \neq 0$

8. With usual notation prove that $\frac{1}{f(D^2)} \cos ax = \frac{1}{f(-a^2)} \cos ax$
where $f(-a^2) \neq 0$

9. with usual notation prove that $\frac{1}{f(D)} xV = x \frac{1}{f(D)} V - \left[\frac{f(D)}{(f(D))^2} \right]$

10. Solve

$$(i) \quad \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = \sin 3x$$

$$(ii) \quad \frac{d^2y}{dx^2} - 8 \frac{dy}{dx} + 9y = 40 \sin 5x$$

$$(iii) \quad (D^2 + 4)y = \cos 2x$$

$$(iv) \quad (D^2 + D + 1)y = \sin 2x$$

$$(v) \quad (D^3 + 2D^2 + 4D + 8)y = x^2$$

$$(vi) \quad (D^2 + 2D + 1)y = x \sin x$$

$$(vii) \quad \frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = x$$

$$(viii) \quad \text{Solve } (D^2 + D)y = x \cos x$$

$$(ix) \quad (D^2 + 1)y = x^2 \sin 2x$$

11. with usual notation prove that $\frac{1}{f(D)} e^{ax} V = e^{ax} \frac{1}{f(D+a)} V$

$$12. \text{ Solve } (i) \quad \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = x^2 e^{3x}$$

$$(ii) \quad \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 4y = e^x \cos x$$

$$(iii) \quad (D^2 + 3D + 2)y = e^{2x} \sin x$$

$$(iv) \quad (D^3 - D^2 + 3D + 5)y = e^x \cos 2x$$

$$(v) \quad (D^2 - 7D - 6)y = e^{2x} (1 + x)$$

$$(vi) \quad (D - 1)^2 y = x \sin x$$

$$(vii) \quad \frac{d^2y}{dx^2} + 4y = x \sin x$$

$$(viii) \quad \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} + y = x \cos x$$

Unit – II

Partial Differential Equations

I. Questions carrying Two Marks each

1. Form the PDE by eliminating arbitrary constants for $z = ax + by + ab$
2. Form the PDE by eliminating arbitrary constants for $z = (x + a)(y + b)$
3. Form the PDE by eliminating arbitrary function from $z = f(x^2 + y^2)$
4. Solve $p + q = \frac{z}{a}$
5. Solve $x^2p + y^2q = z^2$
6. Define (a) Complete solution (b) Singular solution.
7. Solve $\sqrt{p} + \sqrt{q} = 1$
8. Solve $pq = e^{z-px-qy}$
9. Solve $z = pq$
10. Solve $pq = xy$
11. Solve $\frac{\partial^2 z}{\partial x^2} - a^2 \frac{\partial^2 z}{\partial y^2} = 0$
12. Solve $(D^2 + 4DD' + 4(D')^2)z = 0$
13. Solve $2r + 5s + 2t = 0$
14. Solve $\frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 6 \frac{\partial^2 z}{\partial y^2} = e^{x+y}$
15. Solve $(D^2 - 4(D')^2)z = \sin(x + y)$

II. Questions carrying Five Marks each

1. Form the PDE by eliminating arbitrary function from

$$z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$$

2. Form the PDE by eliminating arbitrary function from

$$z = f(x + ay) + g(x - ay)$$

3. Derive the PDE of the form $Pp + Qq = R$ by eliminating arbitrary function ϕ from $\phi(u, v) = 0$.

4. Form the PDE by eliminating arbitrary function from $f(x^2 + y^2 + z^2, xyz) = 0$

5. Explain the method of solving for linear partial differential equation of the first order of the form $Pp + Qq = R$ where P, Q, R are functions of x, y, z

6. Solve $zp - zq = z^2 + (x + y)^2$

7. Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$

8. Solve $x^2p^2 + y^2q^2 = z^2$.

9. Find the complete and singular solution for $z = px + qy + p^2 + q^2$

10. Solve $z^2(p^2 + q^2 + 1) = 1$.

11. Solve $p^2 - q^2 = x - y$

12. Solve $p^2 + q^2 = z^2(x + y)$: use $w = \log z$

13. Explain Charpit's method for solving partial differential equation

$f(x, y, z, p, q) = 0$

14. Solve $z = px + qy + pq$ by Charpit's method

15. Solve $(p^2 + q^2)y = qz$ by Charpit's method

16. Solve $p(q^2 + 1) + (b - z)q = 0$ by Charpit's method

17. Solve $\frac{\partial^2 z}{\partial x^2} - 3 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = e^{2x+3y}$

18. Solve $(D^2 - 5DD' + 4(D')^2)z = \sin(4x - y)$

19. Solve $(D^2 - 6DD' + 9(D')^2)z = 36xy$

20. Solve $r - 2s + t = \tan(y + x)$

Unit – III

Laplace Transforms

I. Question carrying TWO marks.

1. Prove that $L[e^{kt}] = \frac{1}{s - k}$

2. Prove that $L[e^{-kt}] = \frac{1}{s + k}$

3. Prove that $L[a^t] = \frac{1}{s - \log a}$

4. Prove that $L[\cos mt] = \frac{s}{s^2 + m^2}$

5. Prove that $L[\sin mt] = \frac{m}{s^2 + m^2}$

6. Prove that $L[\cosh mt] = \frac{s}{s^2 - m^2}$

7. Prove that $L[\sinh mt] = \frac{m}{s^2 - m^2}$

8. If $L[f(t)] = F(s)$ then prove that $L[e^{at} f(t)] = F(s - a)$

(First shifting property)

9. Find the laplace transform of

- | | | | |
|--------------------------|-------------------------|-------------------------|--------------------------|
| (i) $\sin^2 t$ | (ii) $\cos^2 t$ | (iii) $\sin^2 2t$ | (iv) $\cos^2 2t$ |
| (v) $\sin^3 t$ | (vi) $\cos^3 t$ | (vii) $\sin 4t \cos 3t$ | (viii) $\cos 5t \cos 2t$ |
| (ix) $4\sin^2 t \cos 2t$ | (x) $3\cos^2 t \cos 3t$ | (xi) $e^{2t} \sin 3t$ | (xii) $e^{-3t} \cos 2t$ |
| (xiii) $e^{-2t} \sin 4t$ | (xiv) $e^{2t} \cos 5t$ | (xv) $e^{2t} t^3$ | (xvi) $e^{-3t} t^2$ |
| (xvii) $e^{4t} t^5$ | (xviii) $e^{-4t} t^3$ | | |

10. Find the inverse laplace transform of

- | | | | |
|---------------------------------|-----------------------------------|-----------------------------|-------------------------------|
| (i) $\frac{2s^2 - 5s + 8}{s^3}$ | (ii) $\frac{3s^2 + 10s - 6}{s^4}$ | (iii) $\frac{s}{(s + 4)^2}$ | (iv) $\frac{s}{(s - 3)^5}$ |
| (v) $\frac{5s + 6}{(s + 1)^2}$ | (vi) $\frac{1}{(s - 4)^3}$ | (vii) $\frac{1}{as - b}$ | (viii) $\frac{1}{(2s + 3)^4}$ |

11. Define unit step or Heviside function and find its Laplace transform.

12. Define direc delta function and find its Laplace transform.

13. Find

(i) $L[t^4 H(t-3)]$ (ii) $L[e^{3t} H(t-3)]$ (iii) $L[t H(t-2)]$ (i) $L\left[\frac{H(t-3)}{t}\right]$

II. Questions carrying Five Marks each

14. Evaluate (i) $\frac{3s+7}{s^2-2s-3}$ (ii) $\frac{1}{s(s+1)(s+2)}$

15. Evaluate (i) $\frac{s+12}{s^2+4s}$ (ii) $\frac{2s+3}{(s-1)(s+2)^2}$

15. Evaluate (i) $\frac{3s+7}{s^2+2s-3}$ (ii) $\frac{2s^2-6s+5}{(s-1)(s^2-5s+6)}$

16. Find the Laplace transform of

(i) $L[t \sin at]$ (ii) $L[t \cos at]$

17. Prove that Laplace Transform of a periodic function $f(t)$ with period T is

$$L[f(t)] = \frac{1}{1 - e^{-sT}} \int_0^T e^{-st} f(t) dt$$

18. Find the Laplace transform of

$$f(t) = \begin{cases} 1 & 0 < t < \frac{a}{2} \\ -1 & \frac{a}{2} < t < a \end{cases} \quad \& \quad f(t+a) = f(t)$$

19. Find the Laplace transform of

$$f(t) = \begin{cases} E & 0 < t < \frac{T}{2} \\ -E & \frac{T}{2} < t < T \end{cases} \quad \& \quad f(t+T) = f(t)$$

20. State and Prove Convolution theorem.

21. Verify convolution theorem for the following pair of functions

(1) $f(t) = 1$ & $g(t) = \sin t$

(2) $f(t) = t$ & $g(t) = e^{2t}$

(3) $f(t) = e^t$ & $g(t) = \cos t$

(4) $f(t) = t$ & $g(t) = \cos t$

(5) $f(t) = \sin t$ & $g(t) = \cos t$

22. Using convolution theorem find

(1) $L^{-1} \left[\frac{1}{(s+2)(s+4)} \right]$

(2) $L^{-1} \left[\frac{1}{s(s^2+1)} \right]$

(3) $L^{-1} \left[\frac{1}{s^2(s-a)} \right]$

(4) $L^{-1} \left[\frac{s}{(s^2+a^2)^2} \right]$

(5) $L^{-1} \left[\frac{1}{(s^2+a^2)^2} \right]$

23. Prove that $L[t^n] = \frac{n!}{s^{n+1}}$ when $n > 0$ an integer.

24. If $L[f(t)] = F(s)$ then prove that

$$L[f^n(t)] = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) \dots \dots \dots f^{(n-1)}(0)$$

25. if $L[f(t)] = F(s)$ then prove that $L \left[\frac{f(t)}{t} \right] = \int_0^\infty F(s) ds$.

26. if $L[f(t)] = F(s)$ then prove that $L \left[\int_0^t f(t) dt \right] = \frac{1}{s} F(s)$

27. Find (i) $L \left[\frac{\sin t}{t} \right]$ (ii) $L \left[\frac{\sin^2 t}{t} \right]$

27. Show that $L \left[\frac{\cos at - \cos bt}{t} \right] = \frac{1}{2} \log \left(\frac{s^2 + b^2}{s^2 + a^2} \right)$

or Evaluate $L \left[\frac{\cos at - \cos bt}{t} \right]$ 28. Evaluate $L \left[\frac{\sin at - \sin bt}{t} \right]$

28. Show that $L \left[\frac{e^{-at} - e^{-bt}}{t} \right] = \log \left(\frac{s+b}{s+a} \right)$ or Evaluate $L \left[\frac{e^{-at} - e^{-bt}}{t} \right]$

29. Evaluate $L \left[\frac{1 - \cos at}{t} \right]$ 30. Evaluate $L \left[\frac{\sin ht}{t} \right]$ 31. Evaluate $L \left[\frac{e^{-t} \sin t}{t} \right]$

32. if $L[f(t)] = F(s)$ then prove that $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} F(s)$

33. Define Heaviside function $H(t-a)$

if $L[f(t)] = F(s)$ then prove that $L[F(t-a)H(t-a)] = e^{-as}f(a)$

Unit IV. FOURIER SERIES & TRANSFORMS

I. Question carrying TWO marks.

- 1. Define a periodic function. Give one example.**
- 2. Define fourier series of periodic function $f(x)$ with period $2l$.**
- 3. Define fourier series of periodic function $f(x)$ with period 2π .**
- 4. Define fourier series in the interval $(C, C + 2l)$ of a periodic function $f(x)$ of period $2l$.**
- 5. State Dirichlet's conditions for Fourier series.**
- 6. Define fourier series and fourier coefficients.**
7. Find a_0 in the fourier expansion of $f(x) = x + x^2$ in $(-\pi, \pi)$
8. find the fourier co-efficient a_0 for the function $f(x) = x - 1$ in $(-\pi, \pi)$
9. Find a_0 in the fourier expansion of $f(x) = e^2$ in $(-\pi, \pi)$
10. find the fourier co-efficient a_0 for the function $f(x) = e^{-ax}$ in $(-\pi, \pi)$
11. Find a_0 in the fourier expansion of $f(x) = e^{-x}$ in $(-l, l)$
12. Find a_0 in the fourier series for $f(x) = x \sin x$ in $(0, 2\pi)$
- 13. Define half – range cosine series**
- 14. Define half – range sine series.**
- 15. Define finite fourier sine transform .**
16. Find finite fourier sine and cosine transforms of $f(x)$ in $(0, l)$
17. Find finite cosine transform of $f(x) = 1 + x$ in $(0, 3)$
18. Find finite fourier sine transform of $f(x) = 1$ in $(0, \pi)$
- 19. State the Fourier transform of $\frac{d}{dx}f(x)$**

II. Question carrying FIVE marks.

1. Obtain fourier series for of $f(x) = e^x$ in $(-\pi, \pi)$
2. Obtain fourier series for of $f(x) = e^{-x}$ in $(-l, l)$
3. Obtain fourier expansion of $f(x) = x - 1$ in $(-\pi, \pi)$
4. Obtain fourier expansion of $f(x) = x$ in $(-\pi, \pi)$

- 5 . Obtain fourier expansion of $f(x) = x$ in $(0, 10)$
- 6 . Obtain fourier series for $f(x) = x^2$ in $0 < x < 3$
- 7 . Find finite fourier series for $f(x) = |x|$ in $(-1, 1)$
- 8 . Obtain fourier series with period 3 of the function $f(x) = 2x - x^2$ in $(0, 3)$
- 9 . Obtain fourier series for the function $f(x) = 2x - 1$ in $-1 < x < 1$.

10 . Obtain fourier expansion of $f(x) = \left(\pi - \frac{x}{2}\right)^2$ in $(0, 2\pi)$

- 11 . Obtain fourier series for of $f(x) = x^2$ in $(-\pi, \pi)$ and show that

$$\frac{\pi^2}{4} = 1 + \frac{1}{4} + \frac{1}{9} + \dots$$

- 12 . Find finite fourier series for $f(x) = |x|$ in $(-l, l)$ and hence prove that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

- 13 . Obtain fourier series for of $f(x) = x$ in $-\pi < x < \pi$ and show that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

- 14 . Obtain fourier series for of $f(x) = x + x^2$ in $(-\pi, \pi)$ and show that

$$\frac{\pi^2}{12} = 1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

- 15 . Obtain fourier series for of

$$f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ \sin x & \text{if } 0 < x < \pi \end{cases}$$

- 16 . Obtain fourier series for

$$f(x) = \begin{cases} 1 - x & \text{if } -\pi < x < 0 \\ 1 + x & \text{if } 0 < x < \pi \end{cases} \quad \text{of period } 2\pi$$

- 17 . Obtain fourier series for of

$$f(x) = \begin{cases} 0 & \text{in } -2 < x < 0 \\ a & \text{in } 0 < x < 2 \end{cases}$$

- 18 . Find the fourier series for the function

$$f(x) = \begin{cases} \pi + x & \text{if } -\pi < x < 0 \\ \pi - x & \text{if } 0 < x < \pi \end{cases}$$

19 . Find the fourier series for the function

$$f(x) = \begin{cases} -\pi & \text{if } -\pi < x \leq 0 \\ 1 & \text{if } 0 < x \leq \pi \end{cases}$$

21 . Obtain the half range cosine series for $f(x) = (x - 1)^2$ where $0 < x < 1$.

22 . Obtain the half range cosine series for $f(x) = \sin x$ in $(0, \pi)$.

23 . Find the half range cosine series for $f(x) = \pi - x$ in $(0, \pi)$.

24 . Obtain sine half range series for $f(x) = x^2$ in $(0, \pi)$.

25 . Find the half range sine series for $f(x) = e^x$ in $0 < x < 1$.

26 . Obtain half range sine and cosine series for $f(x) = (x - 1)^2$ in $(0, 1)$.

27 . Find the finite sine transform for $f(x) = \pi x - x^2$ in $(0, \pi)$.

28. Find fourier finite cosine transform of $f(x) = 2 - x$ in $(0, 2)$

29. Find fourier finite cosine transform of $f(x) = 1 + x$ in $(0, 3)$

30 . Find the finite sine and cosine transform for $f(x) = 2x - x^2$ in $(0, \pi)$.

31. Find the Fourier transform of $f(x) = e^{-a|x|}$ $a > 0$.

32. Prove that $\mathcal{F}\left(\frac{d}{dx}f(x)\right) = i\omega f(\omega)$

33. Prove that $\mathcal{F}\left(\frac{d^2}{dx^2}f(x)\right) = -\omega^2 f(\omega)$